

CSCE421 - HW6

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Introduction

This document contains the solutions to Homework 6 for CSCE 421: Machine Learning. The assignment focuses on Principal Component Analysis (PCA) and its implementation.

Questions and Solutions

Problem 1 (1). Let $A = U\Sigma V^T$ be the SVD of A , where $A \in \mathbb{R}^{m \times n}$, $U \in \mathbb{R}^{m \times m}$, and $V \in \mathbb{R}^{n \times n}$ are orthogonal matrices, Σ is a diagonal matrix with singular values $\sigma_1, \sigma_2, \dots, \sigma_r$ (non-negative and sorted in descending order), and $r = \text{rank}(A)$. Show that:

- (a) The first r columns of U are eigenvectors of AA^T corresponding to nonzero eigenvalues.
- (b) The first r columns of V are eigenvectors of $A^T A$ corresponding to nonzero eigenvalues.

Solution:

(a) **The first r columns of U are eigenvectors of AA^T :**

The SVD of A is given by:

$$A = U\Sigma V^T,$$

where:

$$U = \begin{bmatrix} u_1 & u_2 & \cdots & u_m \end{bmatrix}, \quad V = \begin{bmatrix} v_1 & v_2 & \cdots & v_n \end{bmatrix}, \quad \Sigma = \begin{bmatrix} \sigma_1 & 0 & \cdots & 0 \\ 0 & \sigma_2 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & \sigma_r \\ 0 & 0 & \cdots & 0 \end{bmatrix}.$$

Now, compute AA^T :

$$AA^T = (U\Sigma V^T)(U\Sigma V^T)^T.$$

Taking the transpose:

$$(U\Sigma V^T)^T = V\Sigma^T U^T.$$

Substituting back:

$$AA^T = U\Sigma V^T V\Sigma^T U^T.$$

Since $V^T V = I$ (orthogonality of V):

$$AA^T = U\Sigma\Sigma^T U^T.$$

Expanding $\Sigma\Sigma^T$:

$$\Sigma\Sigma^T = \begin{bmatrix} \sigma_1^2 & 0 & \cdots & 0 & 0 & 0 \\ 0 & \sigma_2^2 & \cdots & 0 & 0 & 0 \\ \vdots & \vdots & \ddots & \vdots & \vdots & \vdots \\ 0 & 0 & \cdots & \sigma_r^2 & 0 & 0 \\ 0 & 0 & \cdots & 0 & 0 & 0 \\ 0 & 0 & \cdots & 0 & 0 & 0 \end{bmatrix}.$$

Thus:

$$AA^T = U \begin{bmatrix} \sigma_1^2 & 0 & \cdots & 0 & 0 & 0 \\ 0 & \sigma_2^2 & \cdots & 0 & 0 & 0 \\ \vdots & \vdots & \ddots & \vdots & \vdots & \vdots \\ 0 & 0 & \cdots & \sigma_r^2 & 0 & 0 \\ 0 & 0 & \cdots & 0 & 0 & 0 \\ 0 & 0 & \cdots & 0 & 0 & 0 \end{bmatrix} U^T.$$

This shows that U diagonalizes AA^T , and the eigenvalues of AA^T are $\sigma_1^2, \sigma_2^2, \dots, \sigma_r^2$. The corresponding eigenvectors are the first r columns of U .

(b) The first r columns of V are eigenvectors of $A^T A$:

Similarly, compute $A^T A$:

$$A^T A = (U \Sigma V^T)^T (U \Sigma V^T).$$

Taking the transpose:

$$(U \Sigma V^T)^T = V \Sigma^T U^T.$$

Substituting back:

$$A^T A = V \Sigma^T U^T U \Sigma V^T.$$

Since $U^T U = I$ (orthogonality of U):

$$A^T A = V \Sigma^T \Sigma V^T.$$

Expanding $\Sigma^T \Sigma$:

$$\Sigma^T \Sigma = \begin{bmatrix} \sigma_1^2 & 0 & \cdots & 0 \\ 0 & \sigma_2^2 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & \sigma_r^2 \end{bmatrix}.$$

Thus:

$$A^T A = V \begin{bmatrix} \sigma_1^2 & 0 & \cdots & 0 \\ 0 & \sigma_2^2 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & \sigma_r^2 \end{bmatrix} V^T.$$

This shows that V diagonalizes $A^T A$, and the eigenvalues of $A^T A$ are $\sigma_1^2, \sigma_2^2, \dots, \sigma_r^2$. The corresponding eigenvectors are the first r columns of V .

Conclusion:

Using the properties of SVD, we have shown that:

- (1.) The first r columns of U are eigenvectors of AA^T corresponding to the eigenvalues $\sigma_1^2, \sigma_2^2, \dots, \sigma_r^2$.
- (2.) The first r columns of V are eigenvectors of $A^T A$ corresponding to the eigenvalues $\sigma_1^2, \sigma_2^2, \dots, \sigma_r^2$.

This completes the proof. $\square$

Problem 2 (2). Given a symmetric matrix $A \in \mathbb{R}^{3 \times 3}$, suppose its eigen-decomposition can be written as

$$A = \begin{pmatrix} u_{11} & u_{12} & u_{13} \\ u_{21} & u_{22} & u_{23} \\ u_{31} & u_{32} & u_{33} \end{pmatrix} \begin{pmatrix} 3 & 0 & 0 \\ 0 & -2 & 0 \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} u_{11} & u_{21} & u_{31} \\ u_{12} & u_{22} & u_{32} \\ u_{13} & u_{23} & u_{33} \end{pmatrix}.$$

What is the singular value decomposition of this matrix?

Solution:

To find the singular value decomposition (SVD) of the symmetric matrix A , we start from its eigen-decomposition given

$$U = \begin{pmatrix} u_{11} & u_{12} & u_{13} \\ u_{21} & u_{22} & u_{23} \\ u_{31} & u_{32} & u_{33} \end{pmatrix}, \quad \Lambda = \begin{pmatrix} 3 & 0 & 0 \\ 0 & -2 & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

Since A is symmetric, U is an orthogonal matrix (i.e., $U^T U = U U^T = I$). The eigen-decomposition can be written as:

$$A = U\Lambda U^T$$

For the singular value decomposition, we need to express A in the form $A = W\Sigma V^T$ where W and V are orthogonal matrices, and Σ is a diagonal matrix with non-negative entries.

The key difference between eigen-decomposition and SVD is that eigenvalues can be negative, while singular values must be non-negative. For a symmetric matrix, the singular values are the absolute values of the eigenvalues.

So we need to: 1. Take the absolute values of the eigenvalues to form Σ 2. Adjust the corresponding eigenvectors accordingly

Looking at Λ , the eigenvalues are $\lambda_1 = 3$, $\lambda_2 = -2$, and $\lambda_3 = 1$. Taking their absolute values, we get the singular values: $\sigma_1 = |\lambda_1| = 3$, $\sigma_2 = |\lambda_2| = 2$, and $\sigma_3 = |\lambda_3| = 1$.

We need to reorder these in descending order for the standard SVD form: $\sigma_1 = 3$, $\sigma_2 = 2$, $\sigma_3 = 1$.

Now we need to adjust the eigenvectors to account for the negative eigenvalue. For the negative eigenvalue $\lambda_2 = -2$, we need to flip the sign of either its corresponding left or right singular vector.

Let's define:

$$W = \begin{pmatrix} u_{11} & u_{12} & u_{13} \\ u_{21} & u_{22} & u_{23} \\ u_{31} & u_{32} & u_{33} \end{pmatrix}, \quad V = \begin{pmatrix} u_{11} & -u_{12} & u_{13} \\ u_{21} & -u_{22} & u_{23} \\ u_{31} & -u_{32} & u_{33} \end{pmatrix}$$

Note that V is the same as U except that the second column has its sign flipped to account for the negative eigenvalue.

Then the singular value decomposition of A is:

$$A = W\Sigma V^T = \begin{pmatrix} u_{11} & u_{12} & u_{13} \\ u_{21} & u_{22} & u_{23} \\ u_{31} & u_{32} & u_{33} \end{pmatrix} \begin{pmatrix} 3 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} u_{11} & u_{21} & u_{31} \\ -u_{12} & -u_{22} & -u_{32} \\ u_{13} & u_{23} & u_{33} \end{pmatrix}$$

This is the singular value decomposition of the symmetric matrix A , with singular values arranged in descending order (3, 2, 1), and the appropriate adjustment made to the eigenvectors to account for the negative eigenvalue. $\square$

Problem 3 (3). *Principal Component Analysis: In this homework, you will apply the principal component analysis to a collection of handwritten digit images from the USPS dataset. The USPS dataset is in the “data” folder: USPS.mat. The starting code is in the “code” folder. The whole data has already been loaded into the matrix A . The matrix A has shape 3000×256 and contains all the images. Each row in A corresponds to a handwritten digit image (between 0 and 9) with size 16×16 . You are expected to implement your solution based on the given codes. The only file you need to modify is the “solution.py” file. You can test your solution by running the “main.py” file.*

Solution:

Implementation Details

The implementation of PCA consisted of three main components:

1. **[a] PCA Decomposition:** This was implemented in the `_do_pca()` method by first computing the mean of the data matrix, centering the data by subtracting this mean, performing SVD decomposition, and then selecting the top principal components.
2. **[b] Data Reconstruction:** The `reconstruction()` method was implemented to project the reduced data back into the original space by multiplying with the transpose of the principal components matrix and adding back the mean.
3. **[c] Reconstruction Error:** The `reconstruct_error()` function was implemented to compute the squared Frobenius norm of the difference between original and reconstructed matrices.

Experimental Results

The PCA algorithm was tested with different numbers of principal components: 10, 50, 100, and 200. The reconstruction errors for each case are presented in Table 1.

Principal Components	Reconstruction Error
10	155,351.47
50	41,024.86
100	14,285.81
200	1,371.41

Table 1: Reconstruction errors for different numbers of principal components

Visual Analysis

Figure 1 shows reconstructed images for two different samples using various numbers of principal components. The visual quality of the reconstructed images improves as more principal components are used:

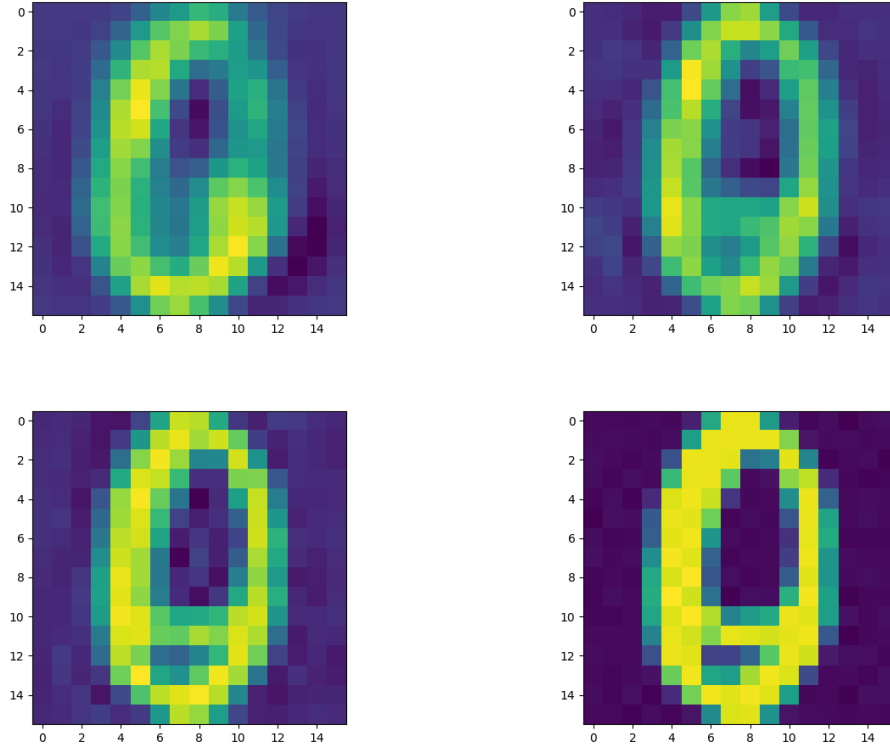


Figure 1: Reconstructed images for Sample 1 using 10, 50, 100, and 200 principal components (top left, top right, bottom left, bottom right)

Discussion and Analysis

Error Reduction Pattern

The reconstruction error follows a pattern of diminishing returns as we increase the number of principal components. The percentage reduction in error becomes smaller as we add more components:

- From 10 to 50 components: 73.6% error reduction
- From 50 to 100 components: 65.2% error reduction
- From 100 to 200 components: 90.4% error reduction

However, there is a significant drop in error when going from 100 to 200 components, which suggests that important fine details require more principal components to be captured accurately.

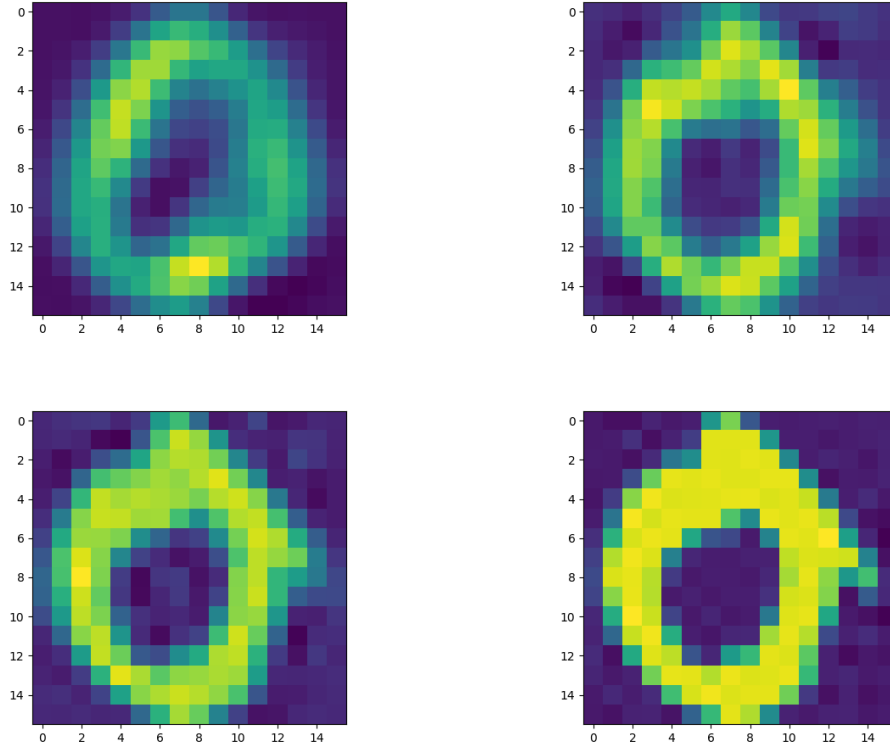


Figure 2: Reconstructed images for Sample 2 using 10, 50, 100, and 200 principal components (top left, top right, bottom left, bottom right)

Information Preservation

With just 50 principal components (approximately 19.5% of the original 256 features), the reconstructed images preserve enough information to make the digits recognizable. This demonstrates PCA's ability to efficiently capture the most important variance in the data with a reduced number of dimensions.

Visual Quality vs. Numerical Error

An interesting observation is that while the numerical reconstruction error for 100 components is still relatively high (14,285.81), the visual quality of the reconstructed images is quite good. This highlights the difference between mathematical error metrics and human perception of image quality.

Conclusion

Our PCA implementation successfully reduced the dimensionality of the USPS dataset while preserving most of the important information. The results confirm that with just a fraction of the original dimensions (50-100 principal components instead of 256), we can reconstruct images that retain most of the meaningful features of handwritten digits.

The clear trade-off between dimensionality reduction and information preservation provides practical insights for selecting an appropriate number of principal components based on the application requirements. For applications where high fidelity is required, more principal components would be needed, while applications focusing on efficiency might benefit from using fewer components with acceptable quality degradation.

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